QOLink Protocol Specification

Version 0.1, 2026-10-28

QOLINK frame:

Preamble	SFD	Header	Seq	Timestamp	Payload	FCS	EFD
Byte 0	1	2..3	4	5..12	13..12+n	13+n..16+n	17+n

Frame Fields:

Preamble: 0x55 (0b01010101, for synchronization help)

SFD (Start Frame Delimiter): Manchester Violation indicating frame start

Header Fields:

Byte[2]..Byte[3]		
class 2b	length 11b	reserved 3b

Class: b00 Low, b01 Medium, b10 High, b11 Credit

Seq (sequence number): 8-bit sequence number, counted separately per class

Timestamp: captured when driver queues data

Payload: variable length

FCS (Frame Check Sequence): CRC-32

Encoding Scheme: Manchester (as defined by IEEE 802.3)